

HAYWARD, CA, USA

WMO: 725850

Lat: 37.654N

Lon: 122.115W

Elev: 13

StdP: 101.17

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 93228

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	4.0	-7.5	2.0	11.6	-4.6	2.6	10.3	9.4	11.6	8.4	12.1	1.5	60	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	9.6	30.8	18.4	28.2	17.8	26.0	17.3	19.7	27.8	18.8	26.2	18.0	24.5	4.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.0	21.7	16.1	11.5	20.8	15.4	11.0	20.3	56.0	27.8	53.2	26.2	50.8	24.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.8	7.2	0.7	35.8	1.1	2.6	0.0	37.6	-0.7	39.1	-1.3	40.6	-2.1	42.5		
			-1.0	21.5	1.3	1.8	-1.9	22.8	-2.6	23.8	-3.3	24.8	-4.3	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	10.5	11.6	13.2	14.2	15.9	17.7	18.8	19.1	19.3	17.2	13.4	10.6
	DBStd	4.04	2.35	2.41	2.66	2.69	2.57	2.78	2.22	2.05	2.84	2.55	2.50	2.53
	HDD10.0	61	22	10	3	1	0	0	0	0	0	0	3	23
	HDD18.3	1341	242	187	161	130	87	43	18	13	18	53	150	240
	CDD10.0	1940	38	56	103	126	183	231	274	282	278	223	104	41
	CDD18.3	177	0	0	2	4	11	25	33	37	46	18	1	0
	CDH23.3	1204	1	2	14	51	108	204	192	176	320	131	6	0
CDH26.7	385	0	0	2	12	36	74	56	53	121	31	1	0	
Wind	WSAvg	3.1	2.3	2.8	3.2	3.7	3.9	3.8	3.7	3.4	3.0	2.5	2.2	2.4
Precipitation	PrecAvg	499	99	91	68	33	17	4	0	1	2	26	55	102
	PrecMax	859	240	323	194	104	67	40	2	20	11	94	167	246
	PrecMin	198	7	0	2	3	0	0	0	0	0	1	4	
	PrecStd	141	67	64	55	26	19	9	0	4	3	25	40	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	22.4	25.2	28.2	30.7	32.9	31.7	32.1	33.7	30.0	23.9	19.3
		MCWB	11.1	13.5	15.6	17.0	17.4	19.1	19.8	19.1	18.8	16.2	14.7	12.8
	2%	DB	17.4	19.2	22.2	24.2	26.4	28.5	27.7	27.5	30.3	27.0	21.2	17.3
		MCWB	11.5	12.7	14.3	15.0	16.5	18.0	18.8	18.7	18.1	16.2	13.8	12.6
	5%	DB	15.9	17.5	19.8	21.1	22.7	25.3	25.1	24.8	27.0	24.3	19.2	16.1
		MCWB	11.7	12.4	13.4	14.2	15.5	17.2	18.0	18.0	17.6	15.4	13.4	12.2
	10%	DB	14.6	15.9	17.8	18.5	20.1	22.6	23.1	23.0	24.2	22.1	17.7	15.0
		MCWB	11.2	11.6	12.7	13.0	14.7	16.3	17.1	17.2	17.0	15.3	13.2	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.3	16.6	18.2	19.2	20.3	20.6	20.5	20.3	19.2	16.9	15.7
		MCDB	16.2	18.5	23.0	26.4	28.0	30.0	29.5	28.6	29.0	25.0	20.7	17.7
	2%	WB	13.4	14.1	15.3	16.1	17.3	18.8	19.1	19.1	19.2	17.8	15.8	14.4
		MCDB	15.4	17.7	20.3	23.0	24.3	27.0	26.8	26.0	26.8	23.3	19.1	16.0
	5%	WB	12.6	13.3	14.3	14.8	15.9	17.7	18.1	18.2	18.3	16.9	15.0	13.5
		MCDB	14.7	16.3	18.3	19.9	21.7	24.7	24.6	24.2	25.3	21.9	17.9	15.2
	10%	WB	11.8	12.4	13.4	13.6	14.9	16.5	17.3	17.4	17.5	16.0	14.1	12.3
		MCDB	13.9	15.1	17.0	17.8	19.7	22.4	22.7	22.6	23.4	20.7	16.9	14.2
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.0	8.4	8.5	8.2	8.9	8.8	8.5	9.6	9.5	8.4	7.7
		MCDBR	9.6	10.8	11.6	13.4	13.8	14.3	12.8	12.1	15.2	13.7	10.7	8.7
		MCWBR	5.6	6.1	6.3	6.9	6.7	6.5	5.7	5.5	7.0	7.0	5.9	5.0
	5% WB	MCDBR	6.8	8.5	9.7	11.7	11.8	12.8	11.8	11.2	12.7	11.0	8.3	5.7
		MCWBR	4.6	5.3	5.7	6.4	6.1	6.1	5.4	5.3	6.0	6.1	5.4	4.1
Clear-Sky Solar Irradiance	taub	0.324	0.331	0.342	0.360	0.367	0.361	0.368	0.349	0.350	0.338	0.325	0.326	
	taud	2.543	2.524	2.480	2.399	2.405	2.426	2.411	2.459	2.445	2.520	2.541	2.544	
	Ebn at Noon	864	904	923	919	914	915	906	918	900	881	855	835	
	Edh at Noon	77	88	101	116	117	115	116	108	103	87	76	73	
All-Sky Solar Radiation	RadAvg	2.39	3.34	4.72	6.16	7.13	7.82	7.43	6.64	5.60	4.12	2.83	2.10	
	RadStd	0.33	0.40	0.42	0.49	0.45	0.32	0.24	0.27	0.21	0.26	0.18	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	-1.17	N/A	N/A	N/A	N/A	-110	+135	N/A

Nomenclature: See separate page